GOVERNMENT ENGINEERING COLLEGE SIWAN

MID TERM EXAMINATION 2025

CSE IOT & VLSI (SECOND SEMESTER)

SUB-CHEMISTRY **TOTAL MARKS 20** **TIME 1.30**

All question have equal marks. Give answer any five question.

1. (a) Elctromagnetic radiation.(b)Photo electric effect.

2. (a) Derived de Broglie equation.

(b) write notes on planks quantum theory.

3.Define following

(a) Ligand (b) Co ordination number (c) Linkage isomerism (d) Write the formula of EDTA.

4.Write the geometrical isomer (cis –trans form) of following.

(a) $[Pt(NH_3)_2Cl_2]$. (b) $[Co(NH_3)_4Cl_2]$.

5.Define following

(a) Hard water (b) soft water (c) hardness of water (d) alkalinity of water.

6. 50 ml of SHW containg 1mg pure $CaCO_3$ consume 20 ml of EDTA. 50ml of water sample consumed 25 ml of same edta solution using EBT .calculate total hardenes of water sample in PPM, ^{0}Cl, ^{0}Fr.

7.A sample water on analyse was found to contain the following in mg/lit

$CaSO_4$=0.45 $MgCO_3$= 0.50 $CaCO_3$= 2.50 Mg SO_4 =0.85 $MgCl_2$=0.80 SiO_2=2.90 NaCl= 2.80

Calculate the amount of lime and soda required to soften 25000 lit water per day for year.if the purity of lime and soda is 90%.

8. Draw molecular orbital diagram of following.

(a) O_2 (b) NO

9.Calculate the bond order and magnetic moment of following.

(a) C_2 (b) He_2

10. Determine hybridisation and geometry .

(a) H_2O (b) PCl_3 (c) I_3^- (d) CO_2